

9 Modbus RTU Function Codes

9.1 Overview

This section describes the function codes needed to upload and download the configuration from a host computer into the instrument.

What's in this section?

The following topics are covered in this section.

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9.2 General Information

This instrument uses a subset of the standard Modbus RTU function codes to provide access to process-related information. Several MODICON function codes are implemented. It is appropriate to define instrument-specific "user-defined" function codes. Where differences occur between the two protocols it will be noted. Several standard Modbus RTU function codes are supported.

Configuration ID Tags

Function codes **20** and **21** use the RS422/485 tag IDs for accessing configuration and process-related data. These tags are fully explained in *Section 10*.

The tag IDs represent the *register addresses* used in the Request Message.

Register Address Structure

Table 9-1 Integer Parameter Type

Register Numbers (Dec)	Name	Access	Notes
1	Type = 1	NOT SUPPORTED	16-bit Unsigned Integer
2	Attribute	NOT SUPPORTED	1 = Read Only, 2 = Read/Write
3	Value (16 bit integer)	Read / Write	
4	Not Used	NOT SUPPORTED	
5	Low Range (16 bit integer)	NOT SUPPORTED	
6	Not Used	NOT SUPPORTED	
7	High Range (16 bit Integer)	NOT SUPPORTED	
8	Not Used	NOT SUPPORTED	
9 to 13	Description Text (ASCII string)	NOT SUPPORTED	

Table 9-2 Floating Point Parameter Type

Register Numbers (Dec)	Name	Access	Notes
1	Type = 2	NOT SUPPORTED	IEEE Floating Point
2	Attribute	NOT SUPPORTED	1 = Read Only, 2 = Read/Write
3	Value (float high word)	Read / Write	
4	Value (float low word)	NOT SUPPORTED	
5	Low Range (float high word)	NOT SUPPORTED	
6	Low Range (float low word)	NOT SUPPORTED	
7	High Range (float high word)	NOT SUPPORTED	
8	High Range (float low word)	NOT SUPPORTED	
9 to 13	Description Text (ASCII string)	NOT SUPPORTED	

Register Count

The register count depends on the data format of the registers being read or written.

Integer data is represented in sixteen bits and is transferred high byte first.

Floating point data is transferred in IEEE 32-bit format.

The register count definitions are:

0001 = Integer Data

0002 = Floating Point Data

9.3 Function Code 20 (14h) - Read Configuration Reference Data

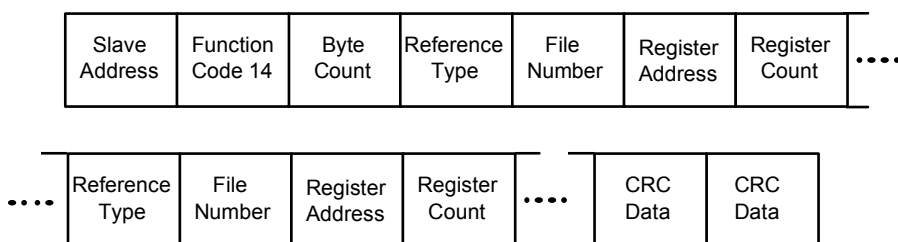
Description

Function code 20 (14 Hex) is used in this instrument to read information stored in its configuration database. Each configuration item is explicitly addressed by a file number and register address. IEEE 32-bit floating point and 16-bit integer formats are supported.

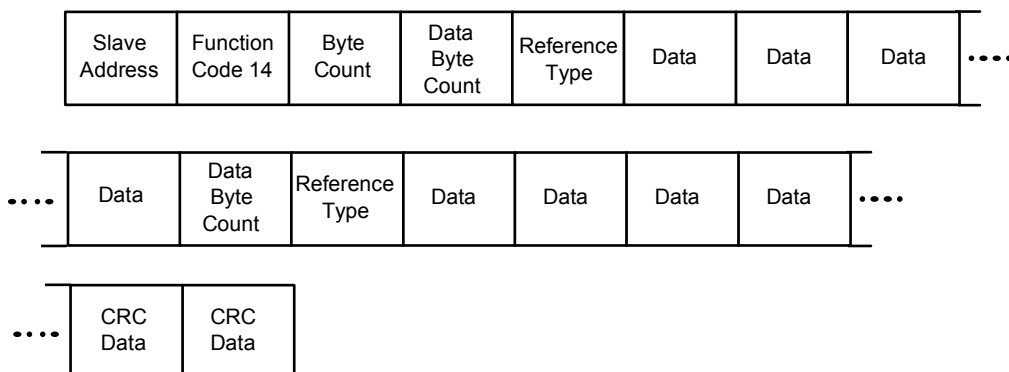
Request and Response Formats

The Request and Response formats for Function code 20 (14 Hex) are shown below. Details for each block reference follow.

Request Message Format



Response Message Format



Byte Count

The Byte Count equals the number of bytes transmitted in either the request or response message and will be the minimum number required to transmit all requested data.

Data Byte Count

The Data Byte Count is the number of data bytes of the *sub response* including the Reference Type but not including itself. A floating point sub response has four bytes of data and one byte representing the reference type making the data byte count equal to five.

Reference Type Definitions

The Reference Type definition is always 06.
See examples in Subsection 9.3.1

File Number

The file number word contains the register number from the register address structure tables on page 3. Although the register address structure tables indicate up to 13 data registers are available for access, only register address 3 is currently supported.

Register Address

The register address word represents the tag ID number for the parameter(s) being accessed. The register address word is made up of two bytes—the MSB = 00 always. The LSB contains the tag ID number. The tag ID numbers represent the parameter's register address(es). See *Section 3* for the tag ID numbers.

Table 9-3 Register Address Format for Function Code 20

Register Address(es) (Decimal)	Register Address(es) (Hex)	Format
001 to 125	0001 to 007D	analog formatted data (2 registers – IEEE 32-bit floating point)
128 to 255	0080 to 00FF	integer formatted data (1 register – 16-bit integer)

9.3.1 Read Configuration Examples

Example #1

The following is an example of a request to read the Gain 1 value using Function code 20.

Request Message (Read (Gain 1) = ID Tag 001)

02 14 07 06 00 03 00 01 00 02 (CRC16)

Where:

02	=	Address
14	=	Function Code 20 (14 hex)
07	=	Byte Count
06	=	Reference Type
00,03	=	File Number (Access Data Value)
00,01	=	Register Address (Standard Access Gain 1 - Tag ID #1)
00 02	=	Register Count (Floating Point Data)
(CRC16)		

This is the response to the above request.

Response Message

02 14 06 05 06 3F C0 00 00 (CRC16)

Where:

02	=	Address
14	=	Function Code 20 (14 Hex)
06	=	Byte Count
05	=	Sub Message Length
06	=	Reference Type (IEEE Floating Point)
3F C0 00 00	=	1.50 (Value of Proportional Band)
(CRC16)		

Example #2

The following is another example of a request and response message using Function code 20.

Request Message (Read LSP #1 = ID Tag 39 and LSP #2 = ID Tag 53)

02 14 0E 06 00 03 00 27 00 02 06 00 03 00 35 00 02 (CRC16)

Where:

02 = Address
 14 = Function Code 20 (14 Hex)
 0E = Byte Count
 06 = Reference Type (IEEE Floating Point)
 00,03 = File Number (Access Data Value)
 00,27 = Register Address (Standard Access LSP #1 - ID Tag 39)
 00,02 = Register Count to read (Floating Point Data)
 06 = Reference Type (IEEE Floating Point)
 00,03 = File Number (Access Data Value)
 00,35 = Register Address (Standard Access LSP #2 - ID Tag 53)
 00,02 = Register Count to read (Floating Point Data)
 (CRC16)

This is the response to the above request.

Response Message

02 14 0C 05 06 43 C8 00 00 05 06 44 60 00 00 (CRC16)

Where:

02 = Address
 14 = Function Code 20 (14 Hex)
 0C = Byte Count
 05 = Data Byte Count (Sub Message Length)
 06 = Reference Type (IEEE Floating Point)
 43 C8 00 00 = 400.0 (Value of Local Setpoint #1)
 05 = Data Byte Count (Sub Message Length)
 06 = Reference Type (IEEE Floating Point)
 44 60 00 00 = 896.0 (Value of Local Setpoint #2)
 (CRC16)

9.4 Function Code 21 (15h) - Write Configuration Reference Data

Introduction

Function Code 21 (15 Hex) is used in this instrument to allow writes of integer and floating point values to the configuration database and override values.

The configuration database of this instrument is located in EEROM. The override values are stored in RAM.

Integer format is used to write to “Digital” configuration items. Floating Point format is used to write to “Analog” configuration items as defined by the configuration ID tags.

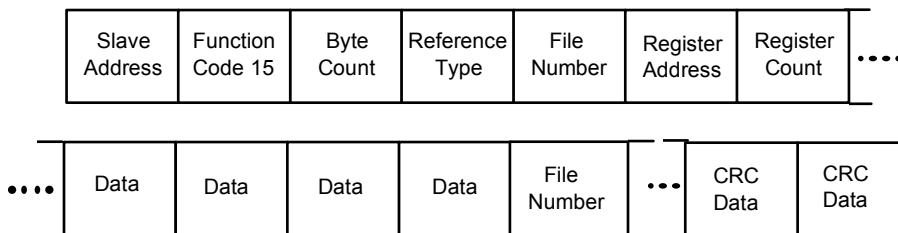
Write Restrictions

Care should be taken not to exceed the 100,000 write limit of the EEROM.

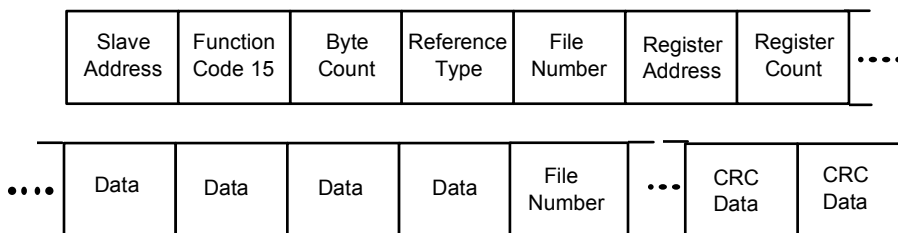
Request and Response Formats

The Request and Response formats for Function code 21 (15 Hex) are shown below. Details for each block reference follow.

Request Message Format



Response Message Format (echo back of request)



The register address is interpreted by this instrument as the tag ID configuration number.

For Infrared Transactions, add three BOFs (C0hex) at the beginning of each message and one EOF (Ffhex) at the end of each message.

Reference Type Definitions

The Reference Type definition is always 06.

See examples in Subsection 9.4.1

File Number

The file number word contains the register number from the register address structure shown in Table 9-1 and Table 9-2. Although the register address structure tables indicate up to 13 data registers are available for access, only register address 3 is currently supported.

Register Address

The register address is used to designate the tag ID number for the parameter being accessed. The register address is made up of two bytes—the MSB = 00 always. The LSB contains the RS422 tag ID number. The tag ID numbers represent the parameter's register address(es). See *Section 10* for the tag ID numbers.

Table 9-4 Register Address Format for Function Code 21

Register Address(es) (Dec)	Register Address(es) (Hex)	Format
001 to 125	0001 to 007D	analog formatted data (2 registers – IEEE 32-bit floating point)
128 to 215 & 255	0080 to 00D7 & 00FF	integer formatted data (2 registers – IEEE 32-bit floating point)

Unrestricted Registers

As mentioned previously, all register data is stored in the EEROM of this instrument with some exceptions. These exceptions were made to allow write access to override information. The registers, which are designated as Override values, are listed below. These registers do not have restrictions on the number of writes.

<i>ID Tag</i>	<i>Register Number</i>	<i>UDC Usage</i>
125	(7Dh)	Computer Setpoint

Restrictions on Parameter Numbers in One Message

The maximum number of writeable parameters per write request is 1.

9.4.1 Write Configuration Examples

Example #1

The following is an example of a request to write the Gain 1 value using Function code 21 (15 Hex).

Request Message (Write Gain 1= 1.5 “ID Tag 1”)

02 15 0B 06 00 03 00 01 00 02 3F C0 00 00 (CRC16)

Where:

02 = Address
15 = Function Code 21 (15 Hex)
0B = Byte Count
06 = Reference Type (IEEE Floating Point)
00 03 = File Number (Access Data Value)
00 01 = Register Address (Standard Access - Gain 1 - ID Tag 1)
00 02 = Register Count (Floating Point Data)
3F C0 00 00 = 1.50
(CRC16)

This is the response to the above request.

Response Message (The response is an echo of the request)

02 15 0B 06 00 01 00 02 00 02 3F C0 00 00 (CRC16)